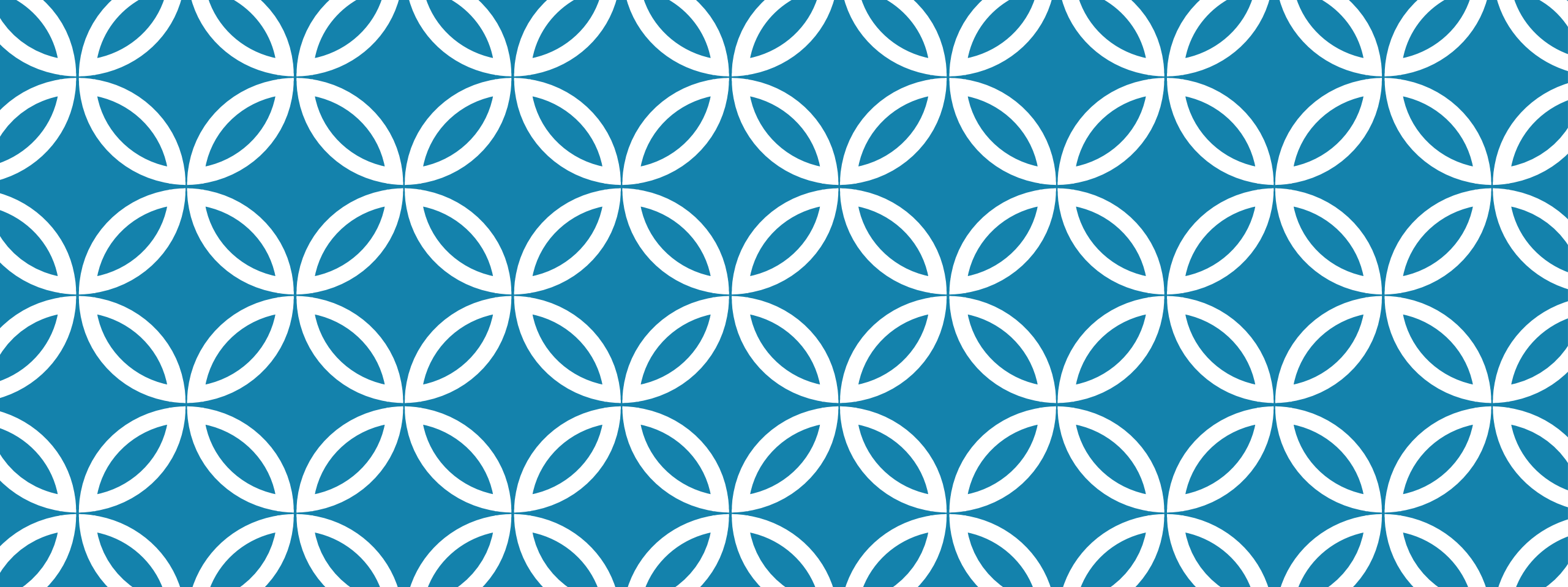




FINAL THEORETICAL REVISION

2024/2025

Module 504 Trauma and Emergency

KEY LABORATORY TESTS THAT SHOULD BE DONE FOR TRAUMA PATIENTS

- **Complete blood count (CBC)**
- **Arterial blood gases (ABG)**
- **Serum glucose**
- **Creatinine**
- **PT, PTT, INR**

MAIN COMPONENTS OF TRAUMA FAST EXAM

Focused Assessment with Sonography in Trauma

- **Cardiac (Subxiphoid)**
- **Right upper quadrant (Hepatorenal)**
- **Left upper quadrant (Splenoarenal)**
- **Suprapubic views**

CHARACTERISTICS OF SEPTIC SHOCK

- **Hypotension**
- **Tachypnoea**
- **Oliguria**
- **Acidosis**

Drug of choice in anaphylactic shock : Adrenaline

SITUATIONS REQUIRE URGENT LAPAROTOMY

- **Hemorrhagic shock + abdominal signs**
- **Sepsis + manifestations of peritonitis**
- **Stab wounds with viscera protrusion**
- **Gun shots**

INDICATIONS FOR SURGICAL INTERVENTION IN CASES OF ABDOMINAL STAB WOUNDS

- **Hemodynamic Instability**
- **Peritoneal Signs: The presence of peritonitis**
- **Evisceration**
- **Confirmed Penetration: peritoneum has been violated.**

CONTRAINDICATIONS FOR EXPLORATORY LAPAROSCOPY IN ABDOMINAL TRAUMA

- **Septic shock.**
- **Multiple injuries.**
- **Significant bleeding.**
- **Immediately life-threatening injuries.**
- **Severe head injury**
- **Severe chest trauma with respiratory failure.**
- **Clinically significant blunt cardiac trauma.**
- **Uncorrected coagulopathy.**
- **Chronic cardiopulmonary disease**

TYPES OF WOUND DRESSINGS

- **Occlusive**
- **Semi-occlusive**
- **Alignate**
- **Osmotic agents (Honey)**

THE LOCAL FACTORS THAT CAN AFFECT WOUND HEALING

- **Vascularity**
- **Irradiation**
- **Immobilization**
- **Tension**
- **Infection**
- **F.B.**
- **Adhesions**
- **↓ Venous drainage**

DIFFERENCES BETWEEN HEALING BY 1RY INTENTION & HEALING BY 2RY INTENTION

- **Less granulation tissue - More granulation tissue**
- **Caused by a clean wound - Caused by an infected wound**
- **Cosmetically accepted scar - Ugly scar**

THE COMPLICATIONS OF RETAINED FOREIGN BODIES

- **Infection & inflammation**
- **Organ dysfunction**
- **Structural damage**
- **Legal problems**

LATE COMPLICATIONS AFTER LIMB AMPUTATION

- **Phantom limb sensation**
- **Stump pain**
- **Joint problems**
- **Skin breakdown**
- **Muscle Weakness and Contractures**
- **Osteoporosis**

CATEGORIES OF BLAST INJURIES AND THEIR CAUSES

- **Primary injury** : Caused by blast wave → over pressure
- **Secondary injury** : Caused by flying debris → shrapnel wounds
- **Tertiary injury** : Cause by blast wind → forceful impact
- **Quaternary injury** ; Caused by other vectors → heat , radiation

KEY COMPONENTS OF DAMAGE CONTROL RESUSCITATION

- **Permissive hypotension and restrictive fluid administration.**
- **Hemostatic resuscitation.**
- **Early hemorrhage control.**
- **Correction of acidosis.**
- **Rewarming.**

THE CAUSES OF CRUSH SYNDROME

- **Drug or alcohol intoxication.**
- **Carbon monoxide poisoning.**
- **Cerebrovascular accident.**
- **Head trauma with coma.**
- **Elderly with hip fracture.**
- **Improper positioning of surgical patient.**
- **Assault with beating.**
- **Pneumatic antishock garment(PASG OR MAST)**

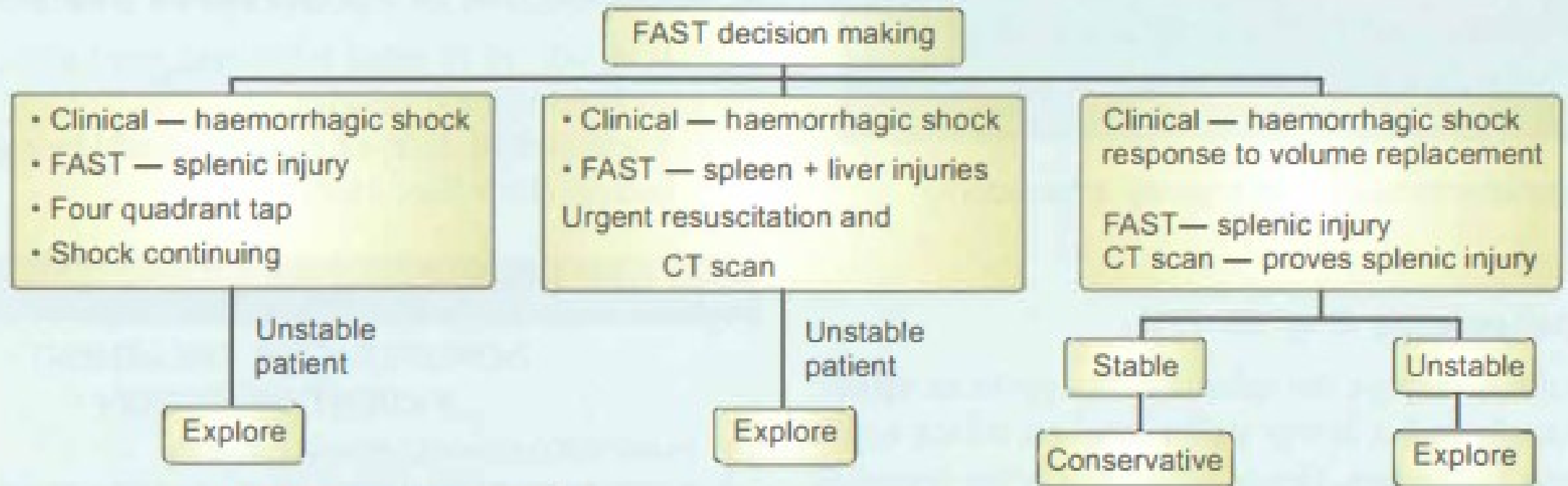
THE CAUSES OF HYPOTHERMIA IN TRAUMA

- **Hemorrhagic shock:** Major blood loss reduces the body's ability to maintain a stable temperature.
- **Environmental exposure:** The injured person is often exposed to cold, wet conditions at the scene for a prolonged period.
- **Medical interventions:** Administration of large volumes of unwarmed intravenous (IV) fluids and blood products during resuscitation can cause core body temperature to drop
- **Altered thermoregulation:** Trauma, especially spinal cord or brain injury, can disrupt the body's natural temperature regulation.
- **Immobilization:** Inability to move due to injury prevents the patient from generating heat through shivering or muscle activity.

THE CHILD ABUSE RED FLAGS

- **Discrepancies in story**
- **Changing history**
- **Inappropriate response**
- **Multiple injuries in past**
- **Classic abuse injuries**
- **Child's development**
- **Sexual abuse**

MANAGEMENT OF SPLENIC INJURIES



FAST: Focussed abdominal sonography in trauma

REQUIREMENTS OF NON-OPERATIVE TREATMENT IN SPLENIC INJURIES

- Haemodynamically stable patients
- No other associated organ injuries which require laparotomy.
- No extra-abdominal injuries which may preclude assessment of abdomen.
- Facilities for emergency resuscitation, ultrasonography, CT scan should be available.
- Patient must be willing for hospitalisation and close observation.
- Regular monitoring by duty doctors, expert nursing staff who maintain hourly vital charts

VACCINATIONS REQUIRED AFTER SPLENECTOMY

- **Pneumococcus**
- **Meningococcus**
- **Haemophilus influenza type B (HiB)**

We may add : Yearly Flu shot and COVID-19

CAUSES OF NON-TRAUMATIC OESOPHAGEAL INJURIES

- **Spontaneous (Boerhaave syndrome)**
- **Iatrogenic injuries (endoscopic)**
- **Severe infection**
- **Corrosive agents**
- **Large foreign bodies as food or bone**

CAUSES OF NON-TRAUMATIC OESOPHAGEAL BLEEDING

- **Mallory-Weiss (severe vomiting or coughing)**
- **Oesophagitis**
- **Oesophageal varices**
- **Oesophageal cancer**

SIGNS FOR POST OPERATIVE LEAK AFTER COLONIC SURGERY

- **Abdominal pain**
- **Tenderness**
- **Fever**
- **Tachypnea**
- **Ileus**
- **Feculant discharge in drain or dressing**

PHARMACOLOGICAL PAIN CONTROL METHODS

Systemic:

- **Paracetamol.**
- **NSAIDs.**
- **Opioids.**

Regional analgesia/ Local Anesthetics.

FACTORS DETERMINING SELECTION OF ANTIBIOTIC AGENT

Antibiotic factors:

- **Empiric therapy: prior to identification of organism, e.g. in critically- ill patients.**
- **Efficacy of the agent: organism's susceptibility to the antibiotic.**
- **Safety of the agent: side effects.**
- **Cost and availability of therapy.**

FACTORS DETERMINING SELECTION OF ANTIBIOTIC AGENT

Patient factors:

- Age.
- Pregnancy and lactation.
- Immune system.
- Comorbidities: Renal and hepatic functions.
- The site of infection and hence tissue penetration, e.g. blood brain barrier

BEST WISHES